In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [36]:
```
import pandas as pd
df=pd.read_csv("C:/Users/DELL/Downloads/city_day.csv")
```

In [37]:
```
print(df.head())
```

```
   City                 Date         PM2.5    PM10      NO       NO2     \
0  Ahmedabad            2015-01-01                         0.92    18.22
1  Ahmedabad            2015-01-02                         0.97    15.69
2  Ahmedabad            2015-01-03                        17.4     19.3
3  Ahmedabad            2015-01-04                         1.7     18.48
4  Ahmedabad            2015-01-05                        22.1     21.42

    NOx      NH3      CO       SO2      O3       Benzene   Toluene   Xylene  \
0    17.15              0.92    27.64   133.36      0.0       0.02     0.0
1    16.46              0.97    24.55    34.06      3.68      5.5      3.77
2    29.7              17.4     29.07    30.7       6.8      16.4      2.25
3    17.97              1.7     18.59    36.08      4.43     10.14     1.0
4    37.76             22.1     39.33    39.31      7.01     18.89     2.78

    AQI     AQI_Bucket
0
1
2
3
4
```

In [38]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype 
---  ------              --------------  ----- 
 0   City                29531 non-null  object
 1    Date               29531 non-null  object
 2    PM2.5              29531 non-null  object
 3    PM10               29531 non-null  object
 4    NO                 29531 non-null  object
 5    NO2                29531 non-null  object
 6    NOx                29531 non-null  object
 7    NH3                29531 non-null  object
 8    CO                 29531 non-null  object
 9    SO2                29531 non-null  object
 10   O3                 29531 non-null  object
 11   Benzene            29531 non-null  object
 12   Toluene            29531 non-null  object
 13   Xylene             29531 non-null  object
 14   AQI                29531 non-null  object
 15   AQI_Bucket         29531 non-null  object
dtypes: object(16)
memory usage: 3.6+ MB
None
```

In [39]:
```
print(df.describe())
```

```
        City                 Date         PM2.5    PM10      NO       NO2     \
count                29531        29531    29531     29531    29531    29531
unique                  26         2009    11717     12572     5777     7405
top     Ahmedabad            2020-06-26
freq                  2009           26     4598     11140     3582     3585

         NOx      NH3      CO       SO2      O3       Benzene   Toluene  \
count     29531    29531    29531    29531    29531     29531     29531
unique     8157     5923     1780     4762     7700      1874      3609
top                         0.0
freq       4185    10328     2328     3854     4022      5623      8041

         Xylene   AQI     AQI_Bucket
count     29531    29531       29531
unique     1562      830           7
top                         Moderate
freq      18109     4681        8829
```

In [40]:
```
df.columns
```

Out[40]:
```
Index(['City              ', ' Date      ', ' PM2.5 ', ' PM10   ', ' NO    ',
       ' NO2   ', ' NOx   ', ' NH3   ', ' CO    ', ' SO2   ', ' O3    ',
       ' Benzene', ' Toluene', ' Xylene', ' AQI   ', ' AQI_Bucket'],
      dtype='object')
```

In [41]:
```
df.columns=df.columns.str.strip()
```

In [42]:
```
df.isnull().sum()
```

Out[42]:
```
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
Xylene        0
AQI           0
AQI_Bucket    0
dtype: int64
```

In [43]:
```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df["AQI_Bucket"]=le.fit_transform(df["AQI_Bucket"].astype(str))
```

In [44]:
```
df["AQI_Bucket"]
```

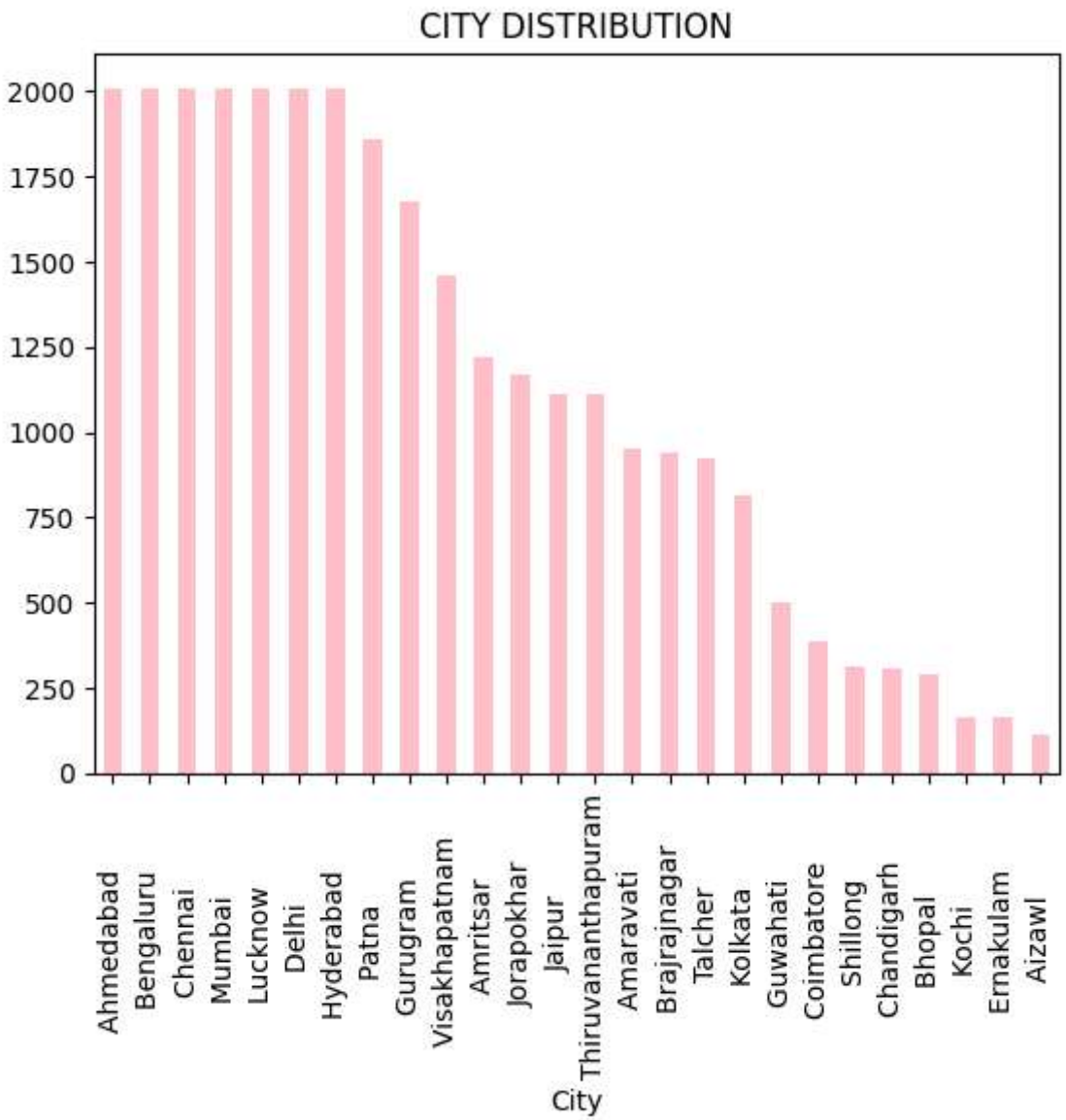
Out[44]:
```
0        0
1        0
2        0
3        0
4        0
        ..
29526    1
29527    4
29528    4
29529    4
29530    1
Name: AQI_Bucket, Length: 29531, dtype: int64
```

In [45]:
```
X=df.drop(["City","Date","AQI_Bucket"],axis=1)
y=df["AQI_Bucket"]
```

In [46]:
```
X=X.apply(pd.to_numeric,errors='coerce')
X=X.fillna(X.mean())
```

In [47]:
```
from sklearn.preprocessing import StandardScaler
SC=StandardScaler()
X_SC=SC.fit_transform(X)
```

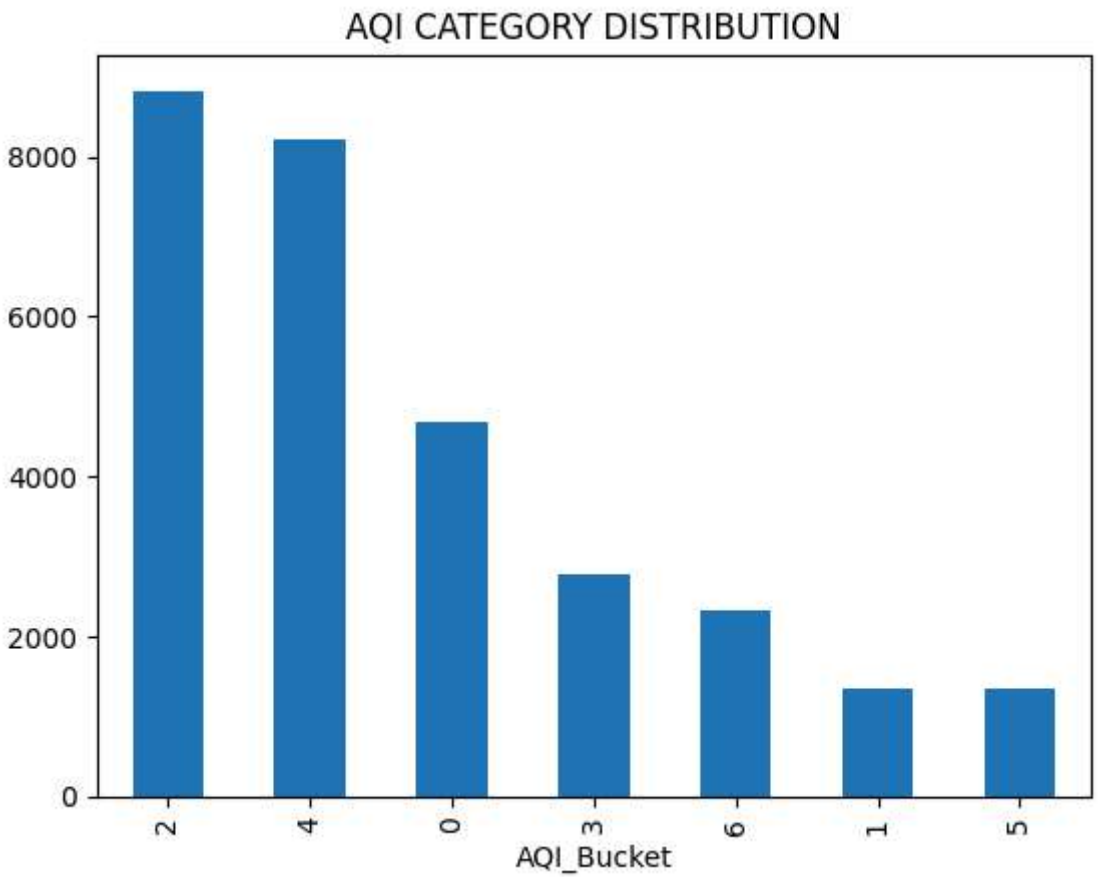
In [48]:
```
import matplotlib.pyplot as plt
df["City"].value_counts().plot(kind='bar',color="pink")
plt.title("CITY DISTRIBUTION")
plt.show()
```



In [49]:
```
df["AQI_Bucket"].value_counts()
```

Out[49]:
```
AQI_Bucket
2    8829
4    8224
0    4681
3    2781
6    2337
1    1341
5    1338
Name: count, dtype: int64
```

In [50]:
```
df["AQI_Bucket"].value_counts().plot(kind='bar')
plt.title("AQI CATEGORY DISTRIBUTION")
plt.show()
```



In [ ]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X_SC, y, test_size=0.2, random_state=42
)
```

In [52]:
```
from sklearn.linear_model import LogisticRegression
lr=LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
```

Out[52]:
```
▾ LogisticRegression
▶ Parameters
```

In [53]:
```
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
```

Out[53]:
```
▾ KNeighborsClassifier
▶ Parameters
```

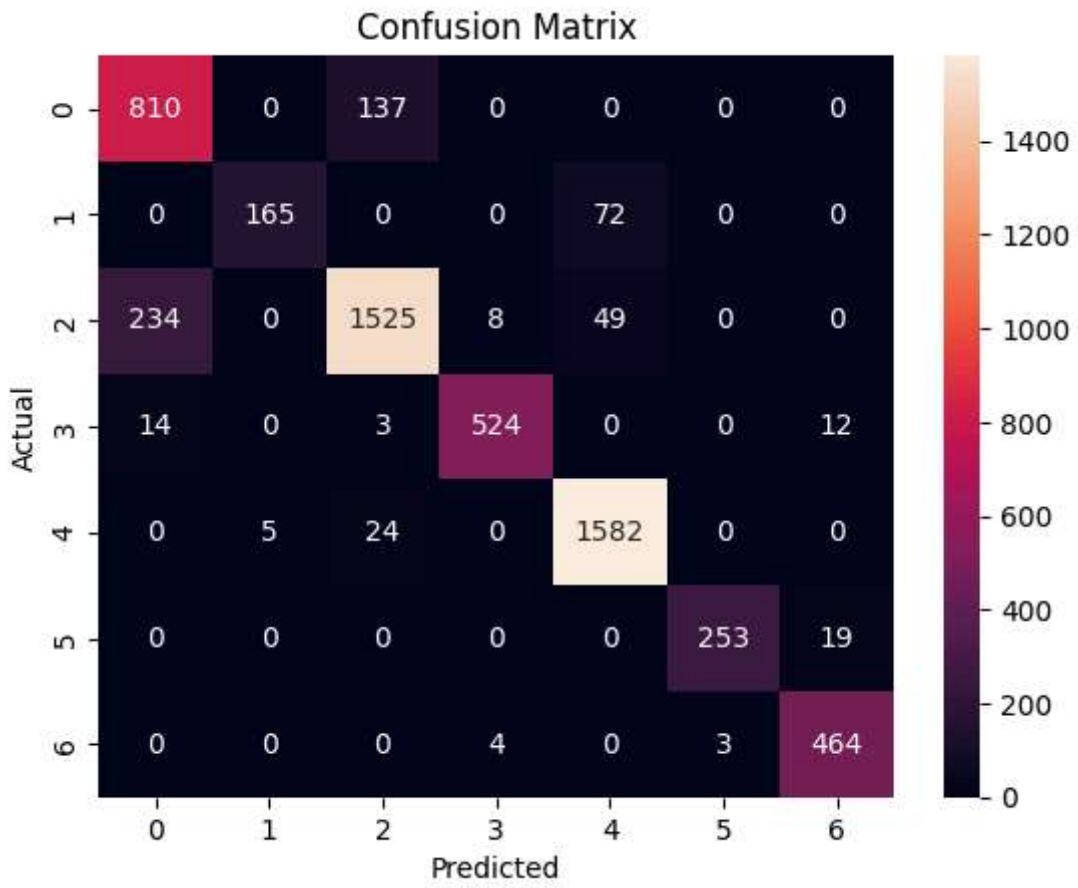
In [54]:
```
y_pred_lr = lr.predict(X_test)
y_pred_knn = knn.predict(X_test)
```

In [55]:
```
from sklearn.metrics import accuracy_score, classification_report
print("LR Accuracy:", accuracy_score(y_test, y_pred_lr))
print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_lr))
```

```
LR Accuracy: 0.9011342475029626
KNN Accuracy: 0.8237684103605891

Classification Report:

              precision    recall  f1-score   support

           0       0.77      0.86      0.81       947
           1       0.97      0.70      0.81       237
           2       0.90      0.84      0.87      1816
           3       0.98      0.95      0.96       553
           4       0.93      0.98      0.95      1611
           5       0.99      0.93      0.96       272
           6       0.94      0.99      0.96       471

    accuracy                           0.90      5907
   macro avg       0.92      0.89      0.90      5907
weighted avg       0.90      0.90      0.90      5907
```

In [56]:

```python
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
cm = confusion_matrix(y_test, y_pred_lr)
sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



In [ ]:
```
sample = X_test[8].reshape(1,-1)
prediction = lr.predict(sample)
print("Predicted:", prediction)
print("Actual:", y_test.iloc[8])
```

```
Predicted: [2]
Actual: 2
```

In [ ]:
```
prediction=lr.predict(sample)
label=le.inverse_transform(prediction)
print("Predicted Category:", label)
```

```
Predicted Category: [' Moderate']
```